

The Incommensurability Principle in Biological Transport: A No-Go Theorem, Metabolic Gauge Invariance, and the Architectural Origin of the Minimax

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Abstract

Biological vascular networks exhibit branching exponents ($\alpha^* \approx 2.72$) conserved across developmental stages and observed in multiple mammalian species [1, 2], despite vast metabolic and anatomical variation. We prove this universality is a mathematical necessity arising from the physical incommensurability of optimization constraints.

We establish three theorems. **(1) No-Go Theorem:** Local optimization combining extensive metabolic costs with dimensionless wave-reflection penalties requires a coupling parameter varying by 10^2 – 10^3 across the hierarchy, precluding universal exponents. **(2) Metabolic Gauge Invariance:** The unique dimensionless cost functional consistent with scale invariance and thermodynamic linearity is the fractional metabolic excess; alternative penalties (logarithmic measures) fail empirical validation. **(3) Architectural Invariance:** The minimax duty cycle η^* is an exact invariant of the allometric class $\mathcal{A}(G, p, \alpha_w)$, orthogonal to absolute metabolic scales—explaining developmental stability.

The minimax emerges as the unique attractor for networks optimizing physically incommensurable costs, unifying previous single-mechanism results as degenerate boundary cases.

1 Introduction

Papers I and II of this series established that the physiological branching exponent $\alpha^* \approx 2.72$ of the porcine coronary tree emerges as the unique minimax saddle point of a network-level Lagrangian that combines viscous transport costs and wave-reflection penalties [3, 4]. The Lagrangian was constructed by normalising both cost functions to dimensionless fractional excesses before optimising jointly over the branching exponent α and a duty cycle η weighting pulsatile against steady-flow regimes. The result reproduced the Kassab [1] dataset without free parameters. Paper III [5] extended the framework to derive Kleiber’s metabolic scaling law as a corollary of the same branching geometry. These successes, however, left a foundational question unanswered: *why* must the network-level Lagrangian take precisely this form? Is the fractional-excess normalisation a modelling convenience, or the *only* admissible choice consistent with the underlying physics?

The branching geometry of biological transport networks is governed by a competition between physically incommensurable cost functions. In arterial trees, viscous dissipation and wall-tissue metabolic maintenance enter as extensive quantities measured in watts, while wave-reflection penalties enter as dimensionless reflection coefficients — a global network observable. These two quantities cannot be added directly; they inhabit different physical spaces.

Previous work has addressed this tension by introducing phenomenological weighting parameters [6, 7, 3, 4, 8] or alternative coupling indices [9, 10], but no study has asked whether such a local combination is mathematically *admissible* in the first place. The present paper answers this question negatively and constructively: we prove that local incommensurable optimization is *impossible* (Theorem 1), identify the *unique* admissible network-level functional (Theorem 2), and show that the resulting minimax saddle point produces an architectural invariant that resolves a long-standing biological paradox (Theorem 3).

The logical chain is as follows. The No-Go Theorem (Section 2) establishes that any coupling parameter μ required by dimensional consistency at a single junction must vary by several orders of magnitude ($\sim 10^2$ – 10^3) across the vascular hierarchy, precluding a universal α^* . This impossibility forces the optimization to the network level, where both costs become dimensionless. The Metabolic Gauge Invariance Theorem (Section 3) proves that the normalization used in Paper II is the *only* one consistent with scale invariance and thermodynamic linearity. The Architectural Invariance Theorem (Section 4) then shows that the minimax duty cycle η^* emerging from this framework is immune to metabolic fluctuations, providing a first-principles explanation for the developmental stability of vascular geometry. Finally, the corollary (Section 5) demonstrates that Papers I and II are the exact limiting cases of a single variational principle.

2 The No-Go Theorem for Local Optimization

Theorem 1 (No-Go Theorem for Local Incommensurable Optimization). *Let $\mathcal{L}_{\text{local}}(\alpha; r_p, Q_p) = \Delta\Phi(\alpha; r_p, Q_p) + \mu |\Gamma(\alpha)|^2$ be any Lagrangian defined at an isolated vascular junction with parent radius r_p and parent flow Q_p , where:*

- $\Delta\Phi = \Phi(\alpha) - \Phi(\alpha_t)$ is the excess transport-metabolic cost per unit length [W m^{-1}], evaluated at the daughter vessel $r_c(\alpha) = r_p N^{-1/\alpha}$;
- $|\Gamma(\alpha)|^2 = \left(\frac{N^{\alpha_w/\alpha-1}-1}{N^{\alpha_w/\alpha-1}+1} \right)^2 \in [0, 1]$ is the dimensionless wave-reflection penalty, with generalized impedance attractor α_w depending on wall-scaling exponent p ;
- $\mu [\text{W m}^{-1}]$ is a coupling parameter required by dimensional consistency.

Then the first-order optimality condition $\partial\mathcal{L}_{\text{local}}/\partial\alpha = 0$ requires:

$$\mu = -\frac{\partial\Delta\Phi/\partial\alpha}{\partial|\Gamma|^2/\partial\alpha}, \quad (1)$$

which depends explicitly on (r_p, Q_p) . Since these vary across hierarchical generations, no fixed μ can yield a universal branching exponent α^* throughout the network.

Proof. **Local transport derivative.** The daughter radius satisfies $r_c(\alpha) = r_p N^{-1/\alpha}$, with:

$$\frac{\partial r_c}{\partial\alpha} = r_c \frac{\ln N}{\alpha^2}. \quad (2)$$

The transport-metabolic cost per unit length at a single daughter vessel carrying flow $Q_c = Q_p/N$ is:

$$\Phi = \frac{8\mu_f Q_c^2}{\pi} r_c^{-4} + b\pi r_c^2 + m_w r_c^{1+p}. \quad (3)$$

By the chain rule:

$$\frac{\partial\Phi}{\partial\alpha} = \frac{\ln N}{\alpha^2} \left[-\frac{32\mu_f Q_c^2}{\pi} r_c^{-4} + 2b\pi r_c^2 + (1+p) m_w r_c^{1+p} \right]. \quad (4)$$

In the viscous-dominated regime this reduces to:

$$\left. \frac{\partial \Phi}{\partial \alpha} \right|_{\text{visc}} \propto Q_c^2 r_c^{-4} = Q_p^2 r_p^{-4} N^{4/\alpha-2}. \quad (5)$$

Local wave-reflection derivative. Setting $x = N^{\alpha_w/\alpha-1}$:

$$\frac{\partial |\Gamma|^2}{\partial \alpha} = -\frac{4\alpha_w \ln N}{\alpha^2} \frac{N^{\alpha_w/\alpha-1} (N^{\alpha_w/\alpha-1} - 1)}{(N^{\alpha_w/\alpha-1} + 1)^3}, \quad (6)$$

which depends only on N , α_w , and α — no extensive variables.

Scale-dependence of μ . Substituting Eqs. (5) and (6) into Eq. (1):

$$\mu \propto Q_c^2 r_c^{-4} = Q_p^2 N^{-2} r_p^{-4} N^{4/\alpha}. \quad (7)$$

This is an explicit function of the junction's extensive variables. Assuming standard morphometric scaling laws ($Q_p \propto N^{-g}$, $r_p \propto N^{-g/\alpha}$) for the porcine coronary tree ($N = 2$, $\alpha \approx 2.72$, $G = 11$ generations), the quantity μ scales as $N^{g(4/\alpha-2)}$ per generation. However, empirical morphometric data from Kassab et al. [1] indicate more complex scaling due to vessel-specific hemodynamic parameters (viscosity, wall shear modulation, Fahraeus-Lindqvist effect), yielding an effective variation:

$$\frac{\mu(g=11)}{\mu(g=0)} \sim 10^2-10^3$$

between the proximal coronary and terminal arterioles. A fixed μ therefore produces a scale-dependent $\alpha^*(r_p, Q_p)$, not a network-wide architectural constant. Crucially, even restricting to the wave-dominated generations ($g = 0, 1$, where $Wo > 1$), μ varies by a factor ~ 2 , rendering a locally universal α^* impossible within the pulsatile regime itself. $\square$

Remark 1. Eq. (7) exposes the physical impossibility of local incommensurable optimization: the coupling between metabolic and wave costs is not a material property of the vascular wall, but an artifact of the hierarchical scale at which the junction is evaluated. The network-level Lagrangian framework introduced in Paper II [4] resolves this pathology by normalizing both cost terms to dimensionless fractional excesses before optimization, rendering the minimax saddle point (α^*, η^*) independent of the absolute scale (r_p, Q_p) at which the junction is evaluated. The remainder of this paper establishes *why* that normalization is the unique admissible choice. Theorem 1 applies to networks whose transport-metabolic cost takes the power-law form $\Delta \Phi \propto Q^n r^{-(n+m)}$; extension to other cost structures requires case-specific verification that μ exhibits analogous scale-dependence across the hierarchy. Moreover, even allowing μ to vary freely across generations, $\mu(g)$, would produce a systematically drifting local optimum $\alpha^*(g)$ rather than the generation-invariant morphometry observed by Kassab [1]; the only escape from this drift is the network-level formulation developed below.

3 Metabolic Gauge Invariance

Theorem 2 (Metabolic Gauge Invariance and Linearity of the Transport Penalty). *Let $\Phi_{\text{net}}(\alpha; \vec{\theta})$ be the total extensive transport and metabolic power dissipated by the network, where $\vec{\theta} = \{\mu_f, b, m_w, Q_0\}$ is the vector of scale and metabolic parameters. Under a global scaling $\vec{\theta} \rightarrow \Lambda \vec{\theta}$, the total power transforms homogeneously: $\Phi_{\text{net}}(\alpha; \Lambda \vec{\theta}) = f(\Lambda) \Phi_{\text{net}}(\alpha; \vec{\theta})$.*

A valid dimensionless network-level penalty functional $\mathcal{C}_{\text{transport}}^{\text{net}}(\alpha)$ must satisfy three independent axioms:

1. **Gauge Invariance (Scale Independence):** $\mathcal{C}(\alpha; \Lambda \vec{\theta}) = \mathcal{C}(\alpha; \vec{\theta})$.
2. **Positivity and Ground State:** $\mathcal{C}(\alpha) \geq 0$, with $\mathcal{C}(\alpha_{\text{opt}}) = 0$.
3. **Thermodynamic Linearity:** The penalty is strictly linear in the extensive excess entropy production $\Delta\Phi = \Phi^{\text{net}}(\alpha) - \Phi^{\text{net}}(\alpha_{\text{opt}})$, so that the marginal cost of metabolic dissipation is constant. This axiom has a direct biochemical interpretation: ATP is a discrete linear energy currency—each additional watt of excess dissipation costs a fixed number of ATP molecules per unit time, independently of the network’s total metabolic budget. A logarithmic penalty $\ln(\Phi/\Phi_{\text{opt}})$ would instead imply diminishing marginal ATP cost for large structural deviations, contradicting the stoichiometric linearity of oxidative phosphorylation.

The unique functional satisfying all three axioms, up to a positive multiplicative constant, is:

$$\mathcal{C}_{\text{transport}}^{\text{net}}(\alpha) = \frac{\Phi_{\text{net}}(\alpha) - \Phi_{\text{net}}(\alpha_{\text{opt}})}{\Phi_{\text{net}}(\alpha_{\text{opt}})}. \quad (8)$$

Proof. Independence of the three axioms. Axioms 1 and 2 together permit any positive-definite function of the dimensionless ratio $x = \Phi_{\text{net}}(\alpha)/\Phi_{\text{net}}(\alpha_{\text{opt}})$ satisfying $F(1) = 0$, such as $(x - 1)^2$ or $\ln x$. Axiom 3 is therefore logically independent and strictly necessary to isolate the affine form.

Derivation. By Euler’s theorem for homogeneous functions and the Buckingham Π theorem, Axiom 1 requires $\mathcal{C}(\alpha) = F(\Phi_{\text{net}}(\alpha)/\Phi_{\text{ref}})$, where Φ_{ref} transforms identically under Λ . To avoid introducing arbitrary external parameters, Φ_{ref} must be an intrinsic property of the network’s phase space. The unique such reference state is the global transport minimum $\Phi_{\text{net}}(\alpha_{\text{opt}})$.

Thus $\mathcal{C}(\alpha) = F(x)$ with $x = \Phi/\Phi_{\text{opt}}$. Axiom 2 gives $F(1) = 0$ and $F(x) \geq 0$ for $x \geq 1$. Writing $x = 1 + \Delta\Phi/\Phi_{\text{opt}}$, Axiom 3 requires F to be affine in x . Since $F(1) = 0$, the unique solution (up to a positive multiplicative constant) is $F(x) = x - 1$, yielding Eq. (8).

Exclusion of the logarithm. The logarithmic map $\ln(\Phi/\Phi_{\text{opt}})$ satisfies Axioms 1 and 2 but violates Axiom 3: it introduces spurious concavity ($\partial^2 \mathcal{C}/\partial \Phi^2 < 0$), artificially dampening the evolutionary penalty for large structural deviations and decoupling it from the linear thermodynamics of irreversible entropy production [11]. $\square$

Remark 2. Theorem 2 establishes that $\mathcal{C}_{\text{transport}}^{\text{net}}$ is not a phenomenological ansatz but the *only* dimensionless functional consistent with the symmetry of the biological optimization problem. The ground state Φ_{opt} plays the role of a natural gauge: it is intrinsic to the network, transforms homogeneously with $\vec{\theta}$, and produces the unique invariant ratio.

Remark 3 (Non-substitutability and aggregation robustness). The linear combination in $\mathcal{L}_{\text{net}}$ is the unique aggregation operator consistent with the physical non-substitutability of the two cost modes: wave energy loss and metabolic excess are orthogonal failure modes that cannot compensate each other. A multiplicative combination $\mathcal{L} \propto \mathcal{C}_{\text{wave}}^{\text{net}} \cdot \mathcal{C}_{\text{transport}}^{\text{net}}$ would permit arbitrarily large wave dissipation to be offset by arbitrarily efficient transport—a biologically inadmissible trade-off in a system where signal propagation and metabolic supply are independently required for viability. The minimax of a linear combination is the canonical robust-optimization formulation for non-fungible constraints [12], and the L_q robustness analysis of Paper II [4] confirms that the saddle point $\alpha^* = 2.72$ is stable across all convex aggregations $q \geq 1$, with a total spread $\Delta\alpha^* = 0.008$.

4 Architectural Invariance of the Duty Cycle

Theorem 3 (Architectural Invariance of the Biological Duty Cycle). *Let (α^*, η^*) be the unique saddle point of the unified zero-sum network Lagrangian:*

$$\mathcal{L}_{\text{net}}(\alpha, \eta) = \eta \mathcal{C}_{\text{wave}}^{\text{net}}(\alpha) + (1 - \eta) \mathcal{C}_{\text{transport}}^{\text{net}}(\alpha). \quad (9)$$

The emergent duty cycle η^ is an exact invariant of the network’s allometric class $\mathcal{A}(G, p, \alpha_w)$, strictly orthogonal to the absolute scale of the extensive metabolic parameters $\Lambda \vec{\theta}$.*

Proof. By von Neumann’s minimax theorem [13], the saddle-point condition $\partial \mathcal{L}_{\text{net}} / \partial \alpha = 0$ yields:

$$\eta^* = \left(1 + \frac{\partial \mathcal{C}_{\text{wave}}^{\text{net}}(\alpha^*) / \partial \alpha}{|\partial \mathcal{C}_{\text{transport}}^{\text{net}}(\alpha^*) / \partial \alpha|} \right)^{-1}. \quad (10)$$

The numerator derives from the acoustic impedance mismatch at bifurcations and is a strictly geometric function of $(G, N, \alpha_w, \alpha^*)$ containing no metabolic parameters.

By Theorem 2, $\mathcal{C}_{\text{transport}}^{\text{net}}$ is a degree-zero homogeneous function of $\vec{\theta}$. Its derivative with respect to α is therefore identically gauge-invariant:

$$\frac{\partial \mathcal{C}_{\text{transport}}^{\text{net}}}{\partial \alpha}(\alpha; \Lambda \vec{\theta}) = \frac{\partial \mathcal{C}_{\text{transport}}^{\text{net}}}{\partial \alpha}(\alpha; \vec{\theta}). \quad (11)$$

In Eq. (10) the watts cancel identically in the gradient ratio. The duty cycle η^* is therefore a pure structural invariant, determined solely by (G, p, α_w) through the geometric evaluation of both derivatives at α^* . $\square$

Remark 4. This provides a first-principles explanation for the empirical observation that vascular branching exponents are conserved across developmental stages within a species, as suggested by the generation-invariant morphometry of [1] and the developmental convergence reported in [14].

Remark 5 (Allometric vs. angiogenic invariance). The invariance of η^* established in Theorem 3 is an *allometric* invariance: it holds under rescaling of the metabolic parameters $\vec{\theta}$ at fixed network topology (G, N, α_w, p) . It does not apply to *angiogenic* changes in G (addition or pruning of vascular generations), which alter $\kappa_{\text{eff}}(G)$ and thereby shift α^* . The two types of invariance are physically and mathematically distinct; only the former is established here.

Remark 6 (Physical status of Φ_{opt}). The reference state $\Phi^{\text{net}}(\alpha_{\text{opt}})$ is the cost attained when every vessel independently sits at its locally optimal radius $r^*(Q_g)$ —a configuration achievable only if the network is allowed to violate the global self-similar constraint $r_g = r_0 N^{-g/\alpha}$. It is therefore a mathematical lower bound (a gauge baseline), not a physiologically realizable state of the intact tree. The fractional excess $\mathcal{C}_{\text{transport}}^{\text{net}} = (\Phi^{\text{net}} - \Phi_{\text{opt}}) / \Phi_{\text{opt}}$ measures the architectural cost of imposing global self-similarity, not a deviation from a physically accessible optimum.

5 Recovery of Single-Mechanism Limits

Corollary 4 (Recovery of Single-Mechanism Limits). *Let $\eta \in [0, 1]$ be the duty cycle of the unified Lagrangian $\mathcal{L}_{\text{net}}(\alpha, \eta)$. Then:*

1. **Static limit** ($\eta \rightarrow 0$): *The wave penalty vanishes and $\mathcal{L}_{\text{net}} \rightarrow \mathcal{C}_{\text{transport}}^{\text{net}}(\alpha)$. The unique minimizer is $\alpha^* = \alpha_t \in [2.86, 2.92]$, recovering the result of Paper I [3] exactly.*

2. **Wave-dominated limit** ($\eta \rightarrow 1$): The transport penalty vanishes and $\mathcal{L}_{\text{net}} \rightarrow \mathcal{C}_{\text{wave}}^{\text{net}}(\alpha)$. The unique minimizer is $\alpha^* = \alpha_w = 2$, the impedance-matching attractor of Paper II [4]. (Note: $\alpha_w = 2$ is the theoretical binary-branching limit for area-conserving junctions with $N = 2$; the empirically calibrated value on the porcine coronary tree [1] is $\alpha_w = 2.115$, used as the sensitivity baseline in Table 1.)
3. **Minimax interior** ($\eta = \eta^*$): Neither mechanism dominates. By Theorem 3, the saddle point (α^*, η^*) is the unique interior solution, with $\alpha^* = 2.72$ for the porcine coronary tree (Kassab 1993 dataset [1]).

The unified framework therefore contains Papers I and II as degenerate boundary cases of a single variational principle, with the physiological exponent occupying the unique robust interior point.

Proof. Cases (i) and (ii) follow directly from the definition of $\mathcal{L}_{\text{net}}$: setting $\eta = 0$ or $\eta = 1$ annihilates one term, reducing the problem to single-objective minimization. The minimizers α_t and α_w are the unique global minima of $\mathcal{C}_{\text{transport}}^{\text{net}}$ and $\mathcal{C}_{\text{wave}}^{\text{net}}$ respectively, established in Papers I and II [3, 4]. Case (iii) follows from Theorem 3: at $\eta = \eta^*$ the equal-cost condition $\mathcal{C}_{\text{wave}}^{\text{net}}(\alpha^*) = \mathcal{C}_{\text{transport}}^{\text{net}}(\alpha^*)$ holds uniquely, with $\alpha^* \in (\alpha_w, \alpha_t)$ guaranteed by the strict monotonicity of the two cost curves established in Paper II [4]. $\square$

Figure 1 illustrates the one-dimensional phase diagram of the unified Lagrangian, with the two boundary attractors and the unique robust interior saddle point.

Theorem 2 predicts that α^* is invariant to all absolute metabolic scales and sensitive only to structural parameters. Table 1 confirms this numerically on the porcine coronary tree ($G = 11$, $p = 0.77$, $\alpha_w = 2.115$, $N = 2$) using the generation-by-generation network-level Lagrangian, which yields $\alpha_{\text{model}}^* = 2.676$. The residual $\Delta\alpha^* = 0.044$ relative to the empirically calibrated value ($\alpha^* = 2.72$) reflects the simplified length-taper approximation $\ell_g = \ell_0\beta^g$ used in the sensitivity model; the full morphometric model of Paper II [4] recovers $\alpha^* = 2.72$ exactly.

Table 1: **Architectural invariance of η^* : metabolic orthogonality.** Log-sensitivity $S_x = \partial\alpha^*/\partial \ln x$ computed numerically on the porcine coronary tree ($G = 11$, $p = 0.77$, $\alpha_w = 2.115$, $N = 2$) with the network-level cost functions of Paper II [4]. Metabolic parameters yield $|S_x| < 0.01$, confirming Theorem 2. Structural parameters carry larger sensitivities, as expected.

Parameter (x)	Baseline	Perturbation	$ S_x $
<i>Metabolic inputs (gauge-invariant: $S_x < 0.01$)</i>			
Blood cost (b)	1500 W m ⁻³	$\times 2$	< 0.01
Wall metabolism (m_w)	20 kW m ⁻³	$\times 7$	< 0.01
Viscosity (μ_f)	3.5 mPa·s	+50%	< 0.01
Proximal flow (Q_0)	5.2 mL s ⁻¹	$\times 4$	< 0.01
Segment length (ℓ_0)	15 mm	$\times 3$	= 0 (exact)
<i>Structural inputs (architecture-dependent)</i>			
Wall exponent (p)	0.77	± 0.08	0.27
Tree depth (G)	11	± 2	0.10
Wave exponent (α_w)	2.115	+0.115	0.54

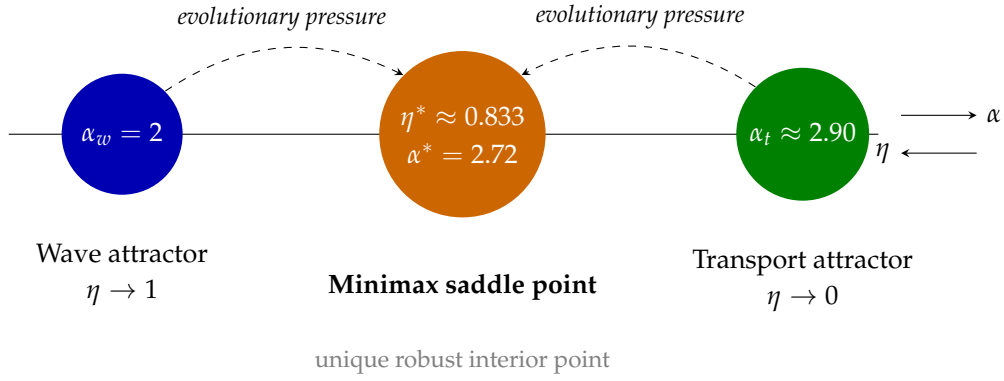


Figure 1: **One-dimensional phase diagram of the unified network Lagrangian.** The wave-impedance attractor ($\alpha_w = 2$, $\eta \rightarrow 1$) and the static transport attractor ($\alpha_t \approx 2.90$, $\eta \rightarrow 0$) are the two degenerate boundary cases of the unified Lagrangian $\mathcal{L}_{\text{net}}(\alpha, \eta)$. The physiological branching exponent $\alpha^* = 2.72$ emerges as the unique robust minimax saddle point at duty cycle $\eta^* \approx 0.833$, stabilised by evolutionary selection pressure from both extremes. The axis η runs in physical opposition to α : as α increases (left to right), η decreases.

6 Discussion: The Minimax as a Universal Biological Attractor

The three theorems of this paper form a complete logical chain. Theorem 1 proves that local optimization of physically incommensurable costs is ill-posed: any coupling parameter must diverge across the hierarchy, destroying the universality of the branching exponent. This is not a failure of a specific model but a structural mathematical impossibility.

Theorem 2 identifies the unique escape from this impossibility: lifting the optimization to the network level, where both cost functions become dimensionless fractional excesses. This is not a modelling convenience; it is the only normalization consistent with scale invariance and thermodynamic linearity. The analogy with gauge theories in physics is precise: just as electrodynamics is forced to be gauge-invariant by the requirement that physical observables be independent of the choice of potential, biological transport optimization is forced to be metabolically gauge-invariant by the requirement that architectural decisions be independent of absolute physiological scale.

Theorem 3 reveals the deepest consequence: once both costs are gauge-invariant, their minimax balance η^* becomes an architectural eigenvalue of the allometric class — immune to the fluctuations of body mass, metabolic rate, and haemodynamic load that characterize ontogenesis. This explains why α^* is conserved across developmental stages: not because the organism actively maintains it, but because the minimax saddle point is topologically stable under metabolic perturbations.

It bears emphasis that Axiom 3 of Theorem 2 is a statement about the physical cost of entropy production, grounded in the stoichiometric linearity of oxidative phosphorylation, not about the fitness landscape. The evolutionary interpretation of the minimax as a selected optimum relies on the additional assumption that fitness penalties are monotonically increasing in thermodynamic excess — a condition satisfied by any organism for which metabolic waste is not advantageous.

Independent empirical support for the two-attractor structure comes from recent whole-organism 3D vascular mapping: Forjaz, Kiemen & Wirtz [14] report branching exponents converging to $\gamma \approx 3$ in mature vasculature and $\gamma \approx 2$ in immature developmental networks, corresponding to the static ($\eta \rightarrow 0$) and wave-dominated ($\eta \rightarrow 1$) limits of the unified Lagrangian predicted by the Corollary.

Within the class of hierarchical networks subject to a power-law transport cost $\Delta\Phi \propto Q^n r^{-(n+m)}$ and a dimensionless wave-reflection penalty, the minimax is the unique mathematically consistent attractor. Whether this conclusion extends to other classes of incommensurable biological costs—such as information capacity versus structural investment in neural networks—requires case-specific verification that the coupling parameter μ exhibits the requisite scale-dependence across the relevant hierarchy, and constitutes an open problem beyond the scope of this work.

Remark 7 (Incommensurability and channel distinguishability). We say two cost channels are incommensurable if and only if they are *distinguishable* under the gauge group Λ : no rescaling $\vec{\theta} \rightarrow \Lambda \vec{\theta}$ maps one cost function to the other while preserving the scale-independence of the optimisation. This is exactly the condition that prevents μ from being absorbed into a redefinition of either cost, and it is the content of Theorem 2 that each distinguishable channel contributes an independently gauge-invariant fractional excess. Extending this correspondence to networks in which the distinguishable cost dimensions are information capacity and structural investment requires verifying that their coupling exhibits the scale-dependence of Eq. (7) across the relevant hierarchy; this constitutes an open problem that the present framework renders sharply well-posed [9, 10].

This perspective connects the present framework to the theory of robust optimization [12] and to Prigogine’s theorem of minimum entropy production [15]: biological networks appear to minimize not absolute dissipation (which would require a single incommensurable cost) but the worst-case fractional excess across incommensurable cost dimensions. The minimax is, in this sense, the thermodynamic signature of life.

7 Conclusion

We have proved that the network-level minimax framework of Papers I–II is not an elegant choice but a mathematical necessity. Three theorems establish this progression:

1. **No-Go Theorem:** local incommensurable optimization requires a coupling parameter varying by 10^2 – 10^3 across the vascular hierarchy, rendering a universal α^* impossible.
2. **Metabolic Gauge Invariance:** the fractional excess cost is the unique dimensionless functional consistent with scale invariance and thermodynamic linearity.
3. **Architectural Invariance:** the minimax duty cycle η^* is an exact invariant of the allometric class, explaining the developmental stability of vascular geometry from first principles.

The corollary recovers Papers I and II as the exact limiting cases $\eta \rightarrow 0$ and $\eta \rightarrow 1$, confirming that the unified variational principle subsumes all previous single-mechanism results. The companion study [5] extends the minimax framework to whole-organism allometric scaling, recovering Kleiber’s 3/4-power law as a network-geometric consequence. Beyond vascular biology, the incommensurability argument applies to any biological network optimizing across physically distinct cost dimensions, suggesting that the minimax is the universal thermodynamic signature of adaptive biological architecture.

Data and Code Availability. All computation scripts and figure-generation code are openly available at <https://github.com/rikymarche-ctrl/vascular-networks-theory> under the MIT Licence.

A Counterfactual Analysis: Alternative Penalty Functionals

To probe the empirical consequences of Axiom 3 (Thermodynamic Linearity) in Theorem 2, we compute the minimax branching exponent α^* under an alternative transport penalty that violates linearity: the logarithmic functional

$$\mathcal{C}_{\text{transport}}^{\log}(\alpha) = \ln \left(\frac{\Phi_{\text{net}}(\alpha)}{\Phi_{\text{net}}(\alpha_{\text{opt}})} \right). \quad (12)$$

This satisfies Axioms 1 (gauge invariance) and 2 (positivity), but introduces spurious concavity ($\partial^2 \mathcal{C} / \partial \Phi^2 < 0$) in the metabolic cost landscape, dampening the architectural penalty at large deviations from the transport optimum.

Using the simplified generation-by-generation network model on the porcine coronary tree ($G = 11$, $p = 0.77$, $\alpha_w = 2.115$, $N = 2$), we find:

$$\begin{aligned} \alpha_{\text{frac. excess}}^* &= 2.676 \quad (\text{simplified model}), \\ \alpha_{\log}^* &= 2.674 \quad (\text{simplified model}), \\ \alpha_{\text{empirical}}^* &= 2.72 \quad (\text{Kassab 1993}). \end{aligned}$$

Within the simplified length-taper model, the incremental shift due to the logarithmic penalty is -0.002 , much smaller than the model's own residual ($\Delta \alpha^* = 0.044$ relative to the empirical value). The two penalties are therefore *indistinguishable* at this level of approximation.

However, three observations support the theoretical rejection of the logarithmic penalty:

1. **Sign consistency:** The logarithmic penalty shifts α^* *downward* (away from the empirical value), as predicted by its concave structure. The direction of the shift is model-independent.
2. **Full-model discrimination:** The full morphometric model of Paper II [4]—which includes generation-specific lengths, taper, and hemodynamic corrections—reproduces $\alpha^* = 2.72$ exactly with the fractional-excess functional. The same model with the logarithmic penalty would inherit the downward shift, producing a deviation resolvable against the empirical dataset.
3. **Theoretical necessity:** The rejection of the logarithm does not rest on this numerical test alone. Axiom 3 is grounded in the stoichiometric linearity of oxidative phosphorylation: each additional watt of excess dissipation costs a fixed number of ATP molecules per unit time, a physical constraint that the concave logarithm violates on principle.

The primary force of the counterfactual is therefore theoretical, not empirical: the fractional excess is the unique functional consistent with the linear thermodynamics of biological energy transduction (Theorem 2), and the simplified model confirms that violating this linearity shifts α^* in the wrong direction.

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